Foundational Models for Geodata Processing

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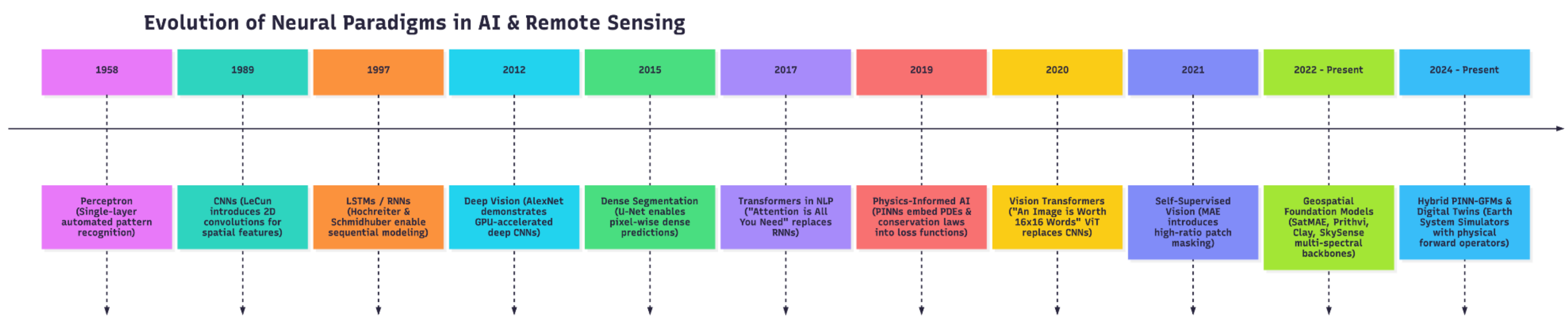
Objectives

- Understanding the Evolution of AI Vision Models
- Learning Approaches of AI models
- Understanding the CNN and ViT based Vision basic difference
- Self Supervise Learning
- A few Geospatila Foundation Models

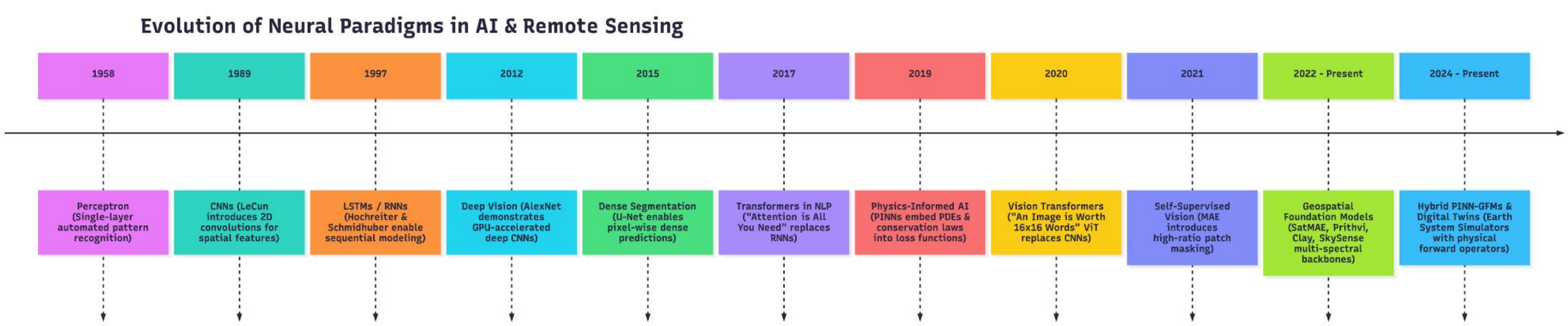
Neural Paradigms in AI, Remote Sensing

- **1958 (Perceptron):** "The Design of an Intelligent Automaton" (Single-layer perceptron introduces automated pattern recognition).
- **1989 (CNNs):** "Backpropagation Applied to Handwritten Zip Code Recognition" (LeCun introduces 2D convolutions for spatial feature extraction).
- **1997 (RNNs/LSTMs):** "Long Short-Term Memory" (Hochreiter & Schmidhuber enable sequential time-series modeling).



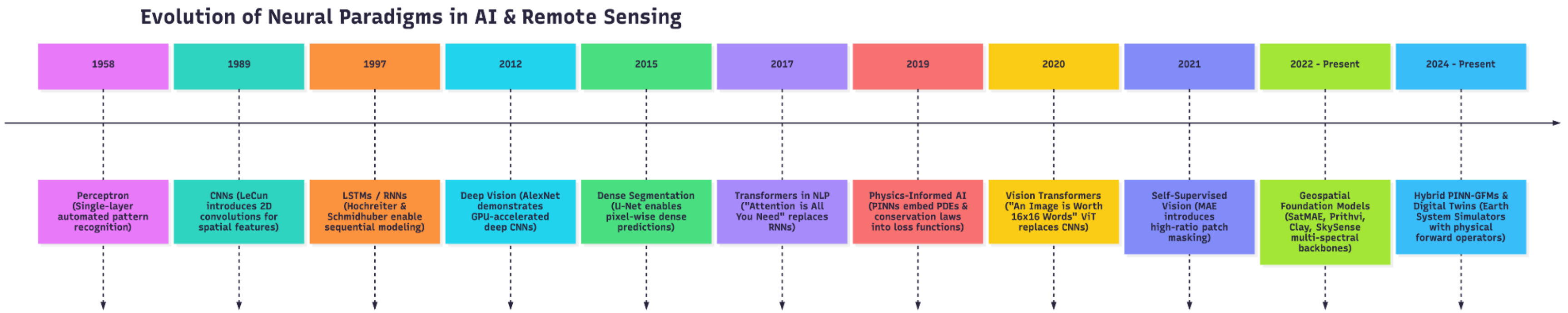
Neural Paradigms in AI, Remote Sensing

- **2012 (Deep Vision):** "ImageNet Classification with Deep CNNs" (AlexNet demonstrates GPU-accelerated deep convolutional networks).
- **2015 (Segmentation):** "U-Net: Convolutional Networks for Biomedical Image Segmentation" (Encoder-decoder architectures enable pixel-wise dense predictions across remote sensing).
- **2017 (NLP):** "Attention is All You Need" (Transformers replace RNNs/LSTMs via self-attention mechanisms).
- **2019 (Physics-Informed AI / PINNs):** "Physics-Informed Neural Networks" (Raissi, Perdikaris & Karniadakis embed partial differential equations [PDEs] and physical conservation laws directly into neural loss functions).



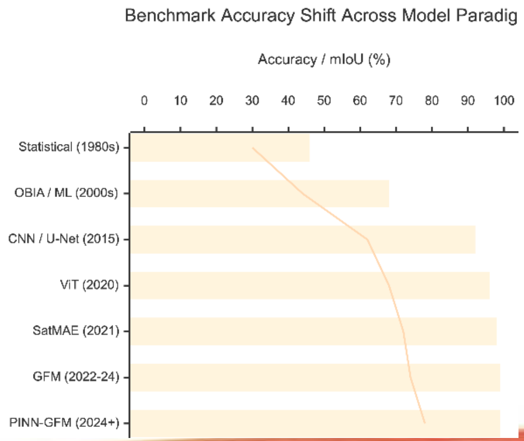
Neural Paradigms in AI, Remote Sensing

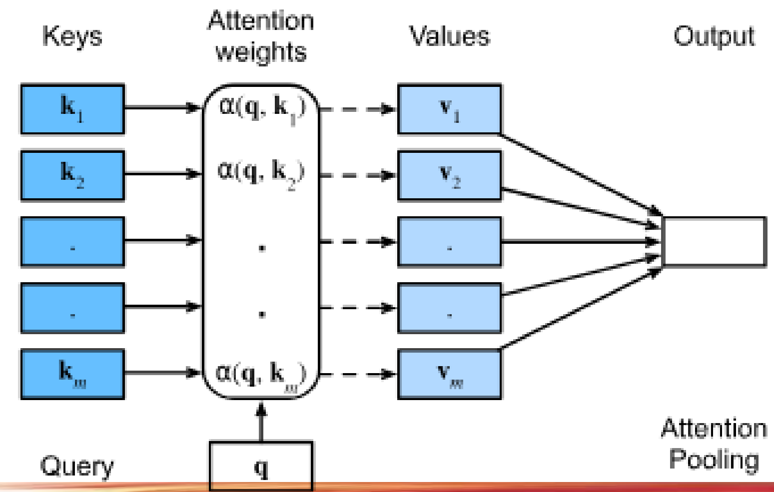
- **2020 (Vision):** "An Image is Worth 16x16 Words" (Vision Transformer [ViT] applies patch-based self-attention to computer vision, replacing standard CNN backbones).
- **2021 (Self-Supervised Vision):** "Masked Autoencoders Are Scalable Vision Learners" (MAE introduces high-ratio patch masking for self-supervised representation learning).
- **2022–Present (Geospatial):** "The Foundation Model Era" (SatMAE, NASA-IBM Prithvi, Clay, SkySense introduce multi-spectral, SAR, and spatial-temporal self-supervised backbones).
- **2024–Present (Multi-Modal Physical Earth AI & Hybrid PINN-GFMs):** "Foundation Models as Earth System Simulators" (Integration of PINN heads, radiative transfer forward operators, and fluid dynamic conservation laws into multi-modal foundation architectures for digital twin



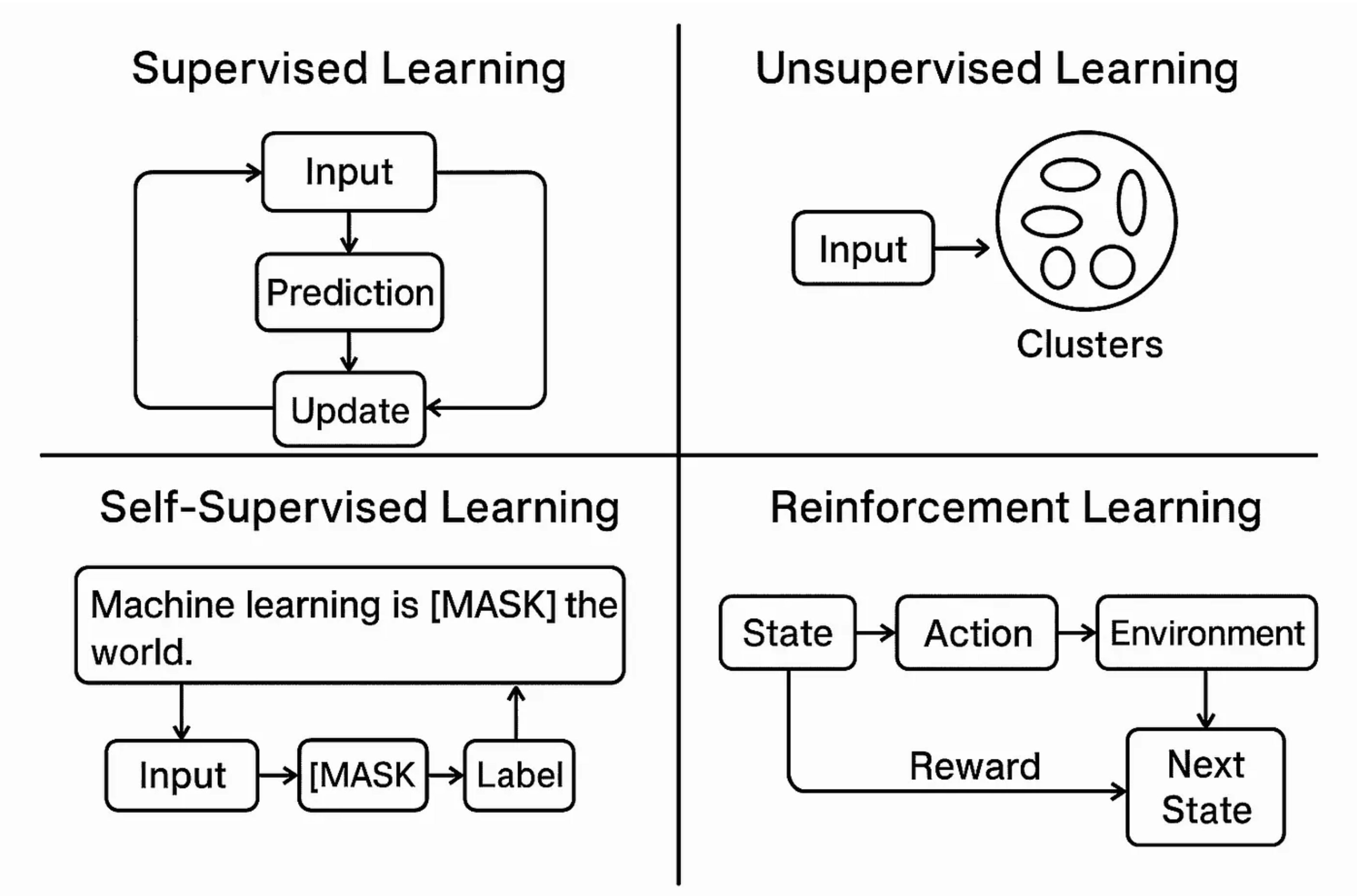
Attention all you need

- 2015 (Segmentation): Convolutional Networks
- 2017 (NLP): "**Attention is All You Need**" (Transformers replace RNNs).
- 2020 (Vision): "An Image is Worth 16x16 Words" (ViT replaces CNNs).
- 2022-Present (Geo): "The Foundation Model Era" (SatMAE, Prithvi, Clay).

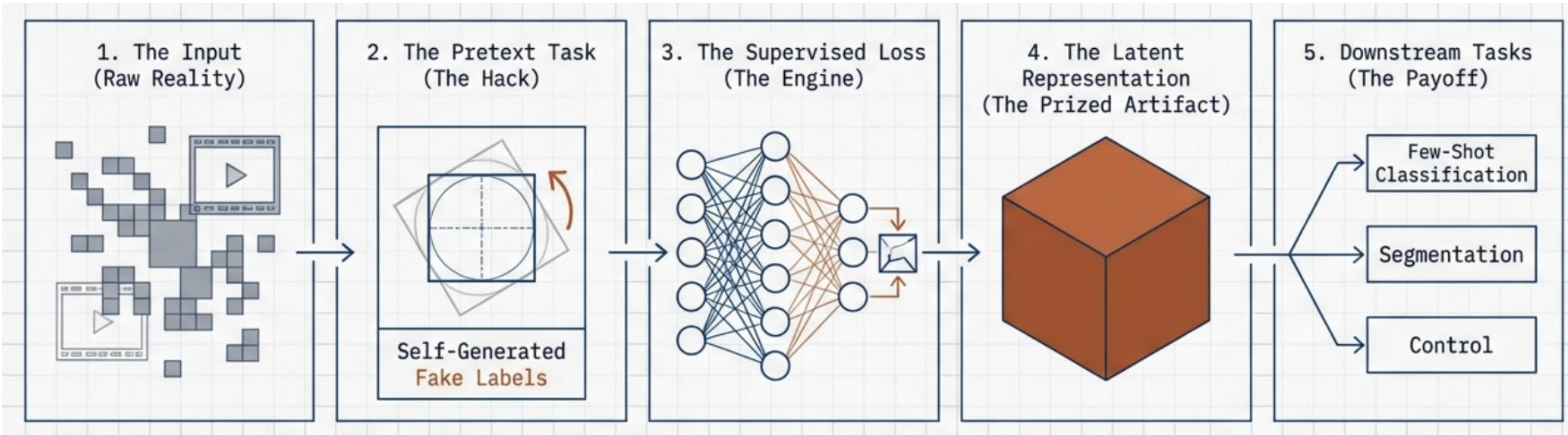


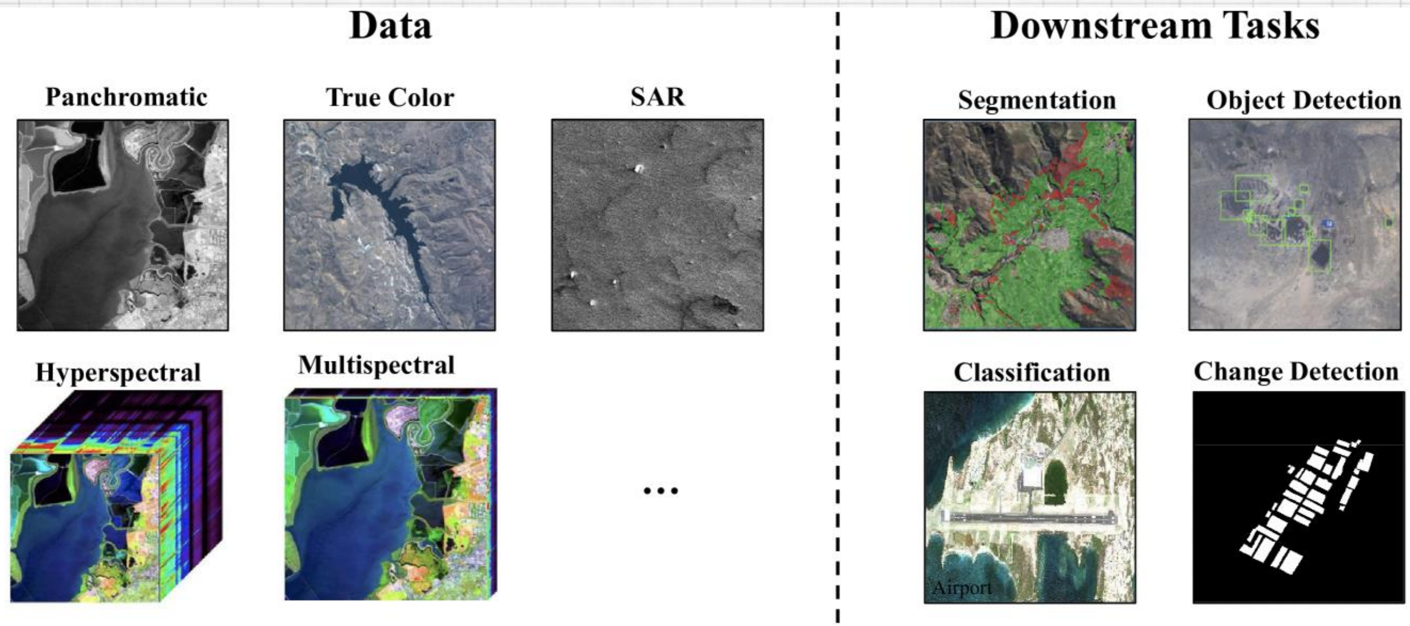


Learning Approaches



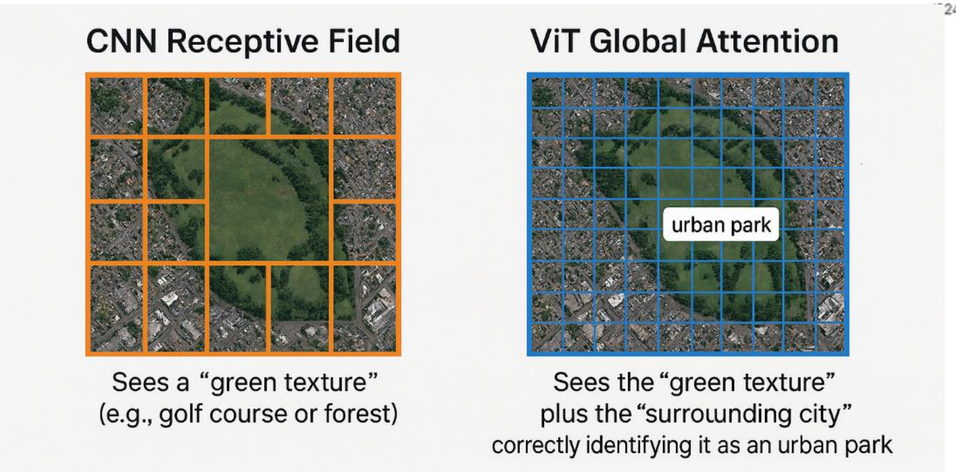
Self Supervised Learning



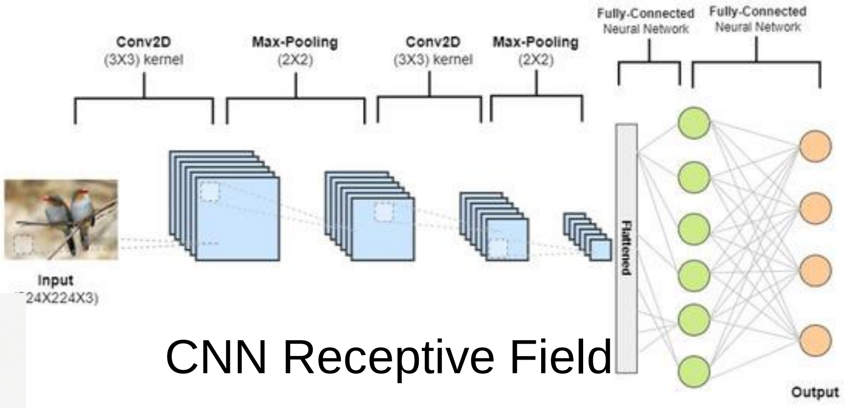


CNN Vs Vit

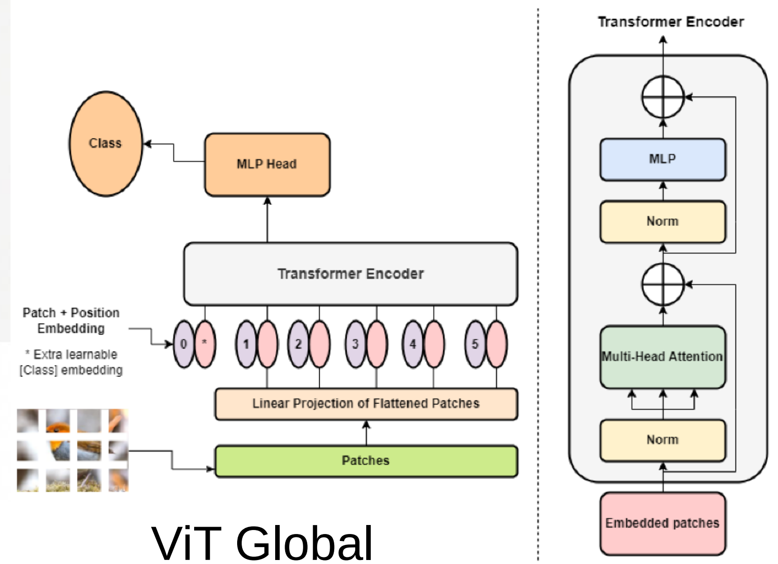
- Long-Range Context
- Input Flexibility



- **Long-Range Context:** A CNN sees a "green texture" (could be a golf course or forest). A VIT sees the "green texture plus the "surrounding city," correctly identifying it as an urban park.
- **Input Flexibility:** CNNs hate changing input sizes. VITs treat pixels as tokens, making it easier to handle different image sizes or missing data (clouds).



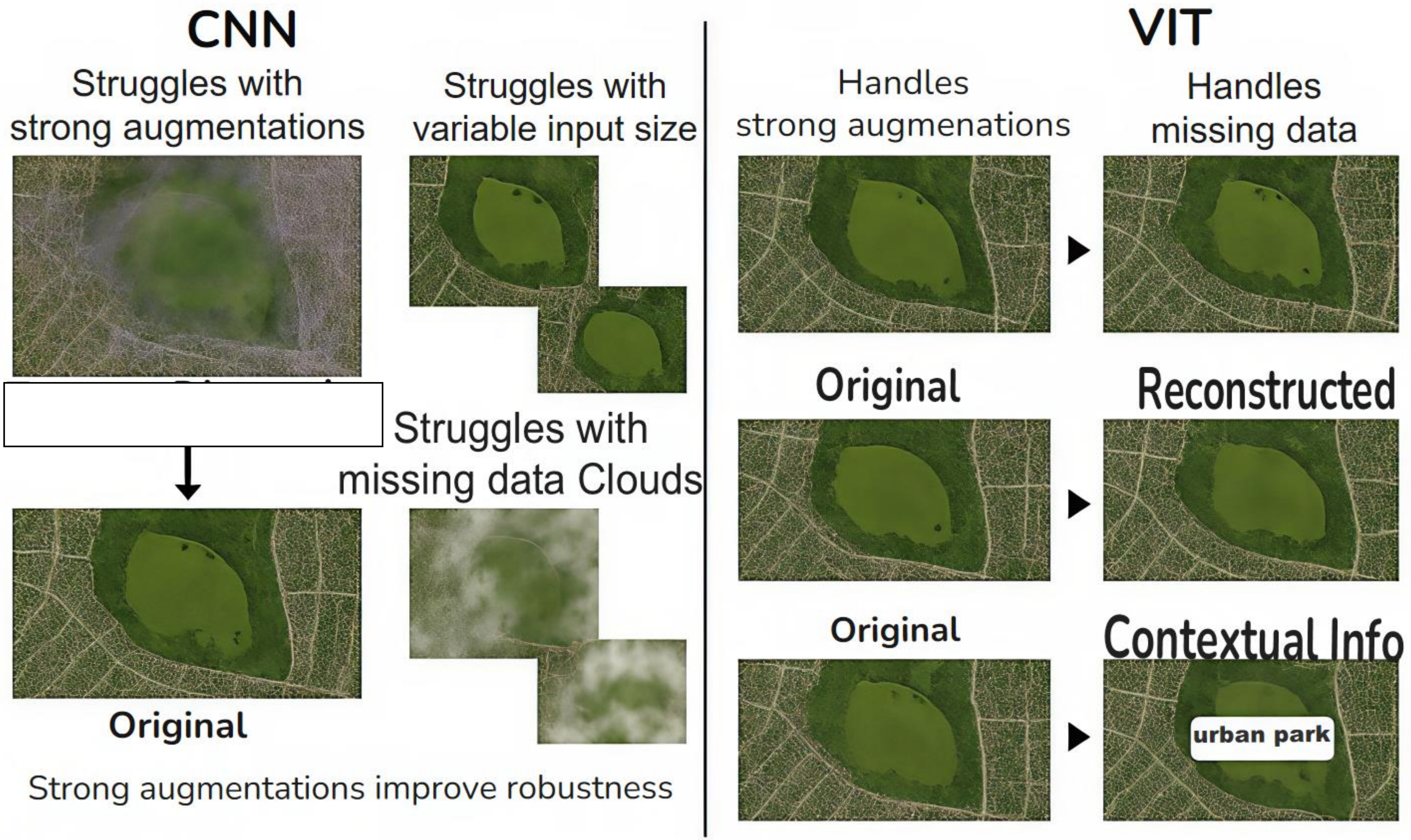
CNN Receptive Field



ViT Global Attention

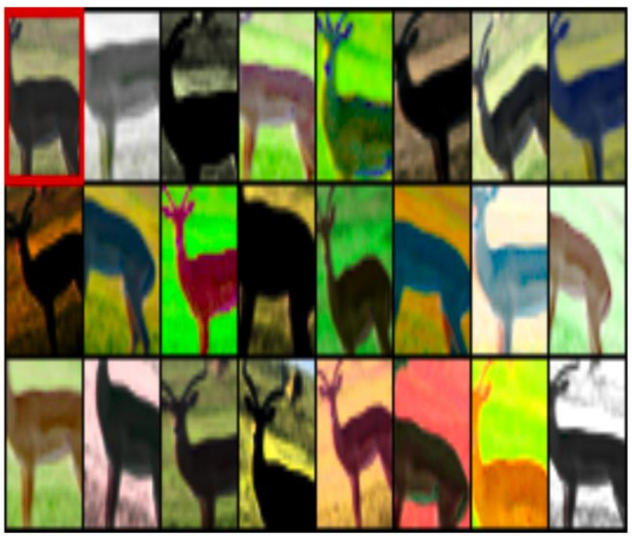
Image Source: https://doi.org/10.3390/app13095521

CNN vs. Vision Transformer (ViT) for Remote Sensing Images

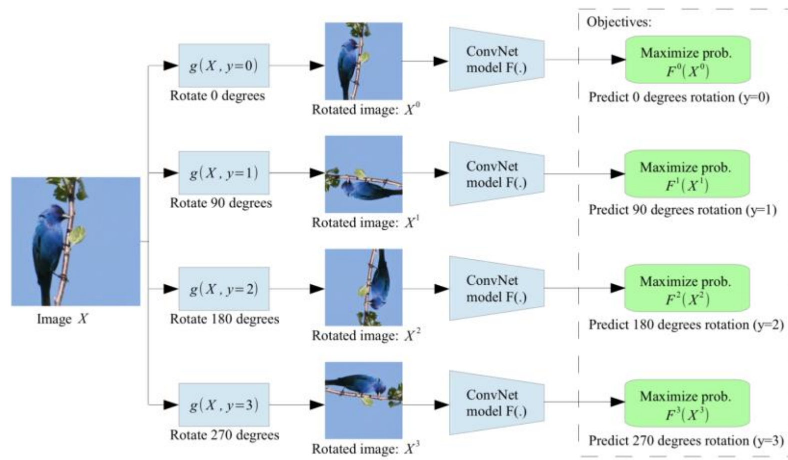


Augmentation Distortion

- **Exemplar-CNN**
  - Creating a sample on similar images (Dosovitskiy et al., 2015)

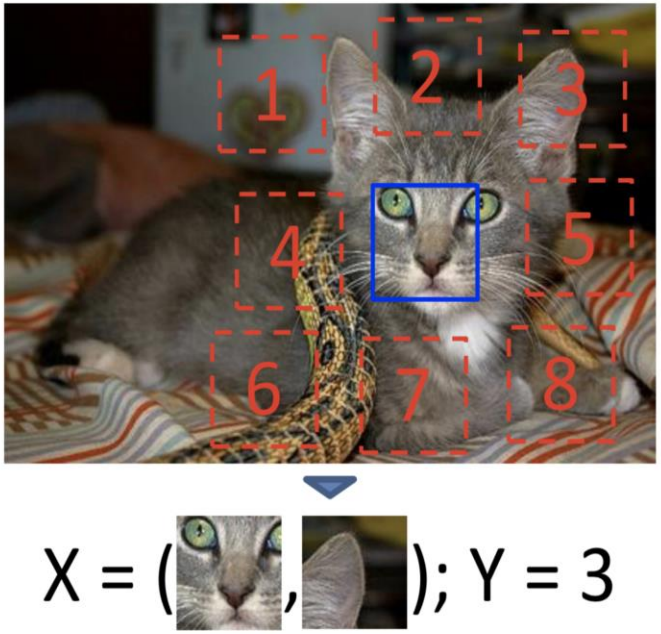


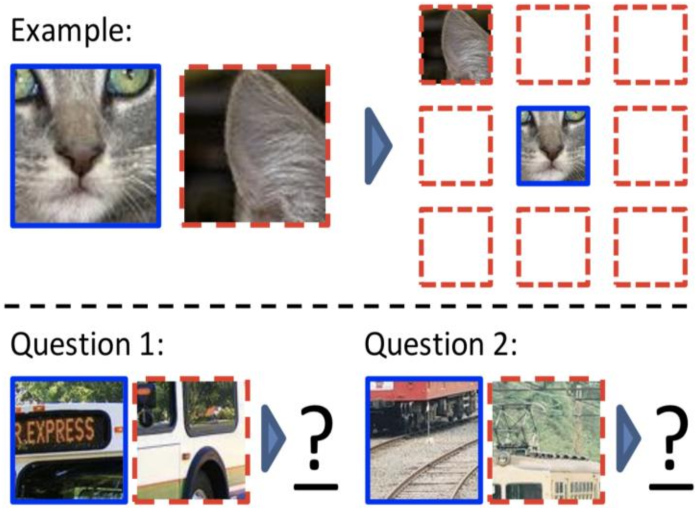
- Rotation
- **Rotation** of an entire image (Gidaris et al. 2018



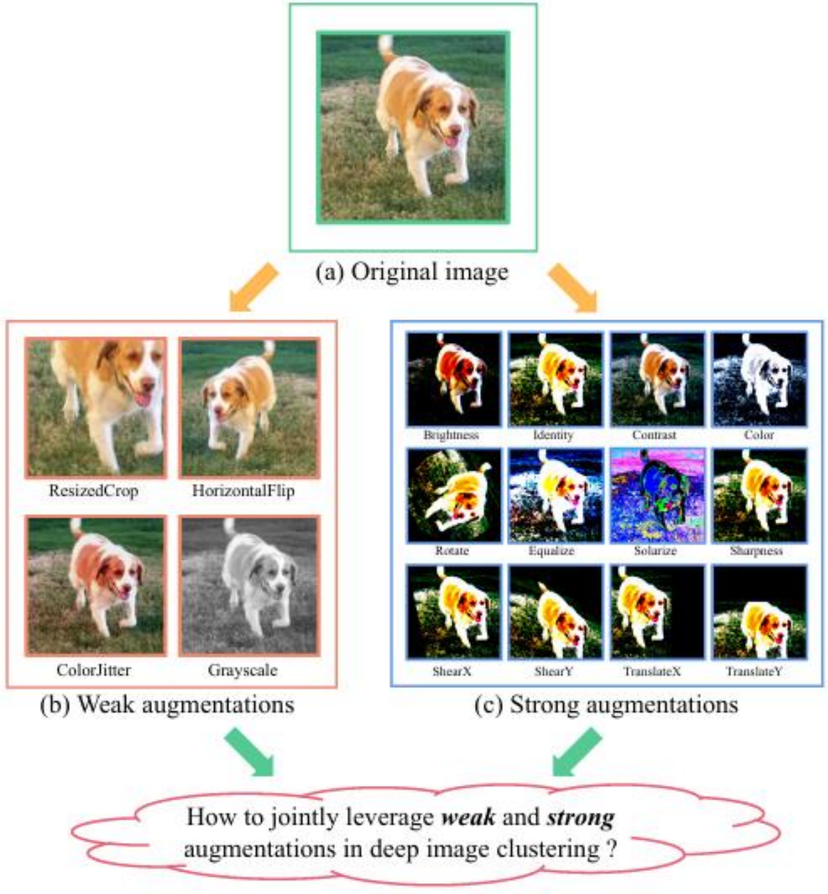
Augmentation
Patches

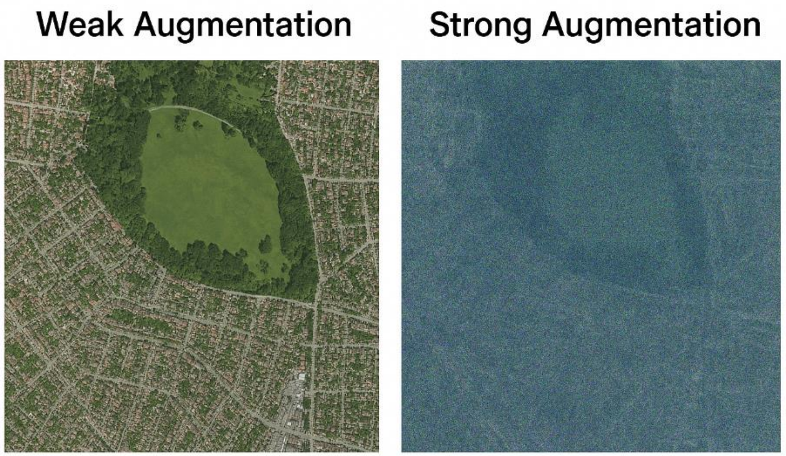
- **Relative position**
  - neighbouring Patch Doersch et al. (2015)
- **Solve Zig saw Puzzle**
  - Noroozi & Favaro, 2016)





Types of Augmentation





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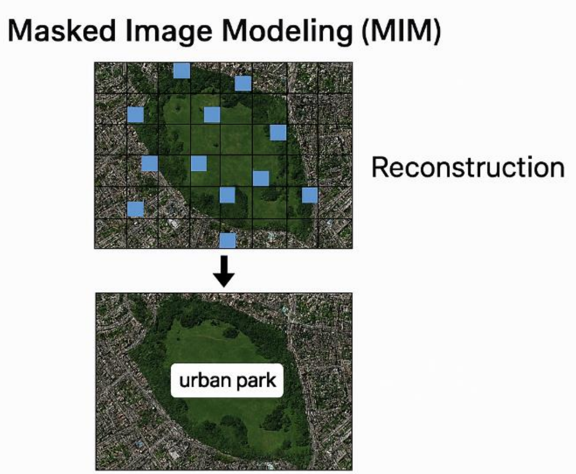
Masked Image Modelling

- **Masking** & **Self-supervised** learning step
- **Reconstruction** process with **Attention** (prediction under the surrounding Global context)
- ViT learns/sees the Semantic "green texture and surrounding city, correctly", and identifies it as an urban park.



Long-Range Context: A ViT sees the "green texture" plus the "surrounnding city," correctly identifying it as an urban park.

Masked Image Modeling: A Survey

Masked Image Modelling (Reconstruction)

Computation:

$P = \{p_i \mid p_i \in \mathbb{R}^{h \times w \times c}\}_{i=1}^{n} \leftarrow \text{split}(X, h, w)$

$P = T_\omega(P)$

$I_v, I_m \leftarrow \text{mask}(P, \alpha, n)$

for $i \in \{1, \ldots, n\}$ **do**

  if $i \in I_v$ **then**

    $H[i] \leftarrow E_\theta(p_i)$

  else if $i \in I_m$ **then**

    $H[i] \leftarrow M$

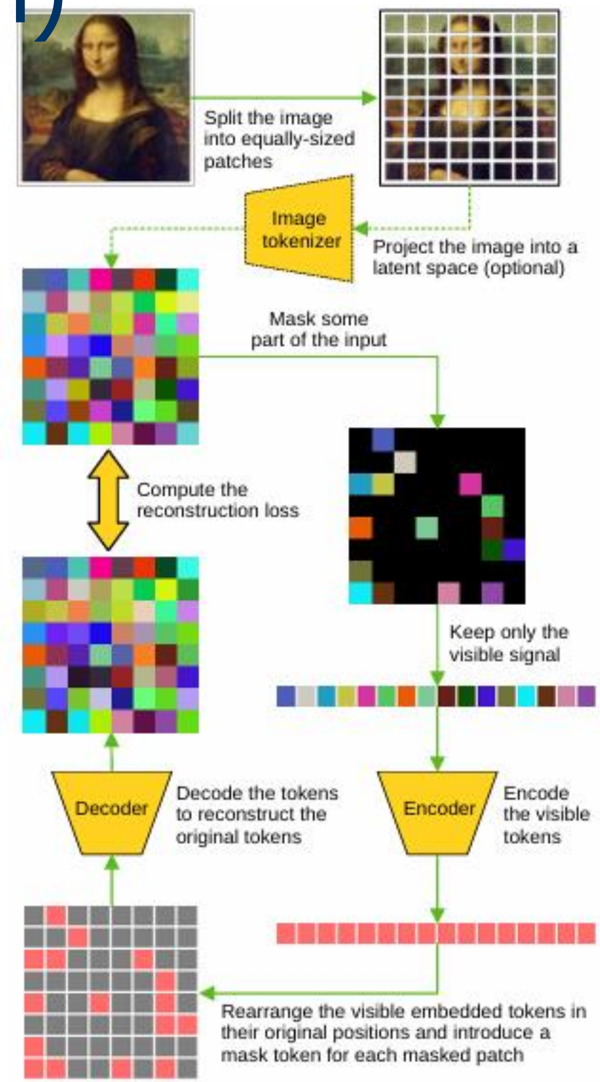
$\hat{P} \leftarrow D_\phi(H)$

$\mathcal{L}(\phi, \varphi, \theta, M) \leftarrow d\left(\hat{P}, P\right)$

$\theta \leftarrow \theta - \eta \cdot \frac{\partial \mathcal{L}}{\partial \theta}$

$\phi \leftarrow \phi - \eta \cdot \frac{\partial \mathcal{L}}{\partial \phi}$

$M \leftarrow M - \eta \cdot \frac{\partial \mathcal{L}}{\partial M}$



Masked Image Modeling: A Survey ,2025

Masked Image Modelling (Contrastive)

Computation:

$\hat{X} \leftarrow \text{augment}(X, \text{'weak'})$

$\tilde{X} \leftarrow \text{augment}(X, \text{'strong'})$

$\hat{P} = \{\hat{p}_i | \hat{p}_i \in \mathbb{R}^{h \times w \times c}\}_{i=1}^{n} \leftarrow \text{split}(\hat{X}, h, w)$

$\tilde{P} = \{\tilde{p}_i | \tilde{p}_i \in \mathbb{R}^{h \times w \times c}\}_{i=1}^{n} \leftarrow \text{split}(\tilde{X}, h, w)$

$I_v, I_m \leftarrow \text{mask}(\tilde{P}, \alpha, n)$

for $i \in \{1, \ldots, n\}$ **do**

  $\hat{H}[i] \leftarrow E_\theta(VP_\varphi(\hat{p}_i))$

  if $i \in I_v$ **then**

    $\tilde{H}[i] \leftarrow E_\theta(VP_\varphi(\tilde{p}_i))$

  else if $i \in I_m$ **then**

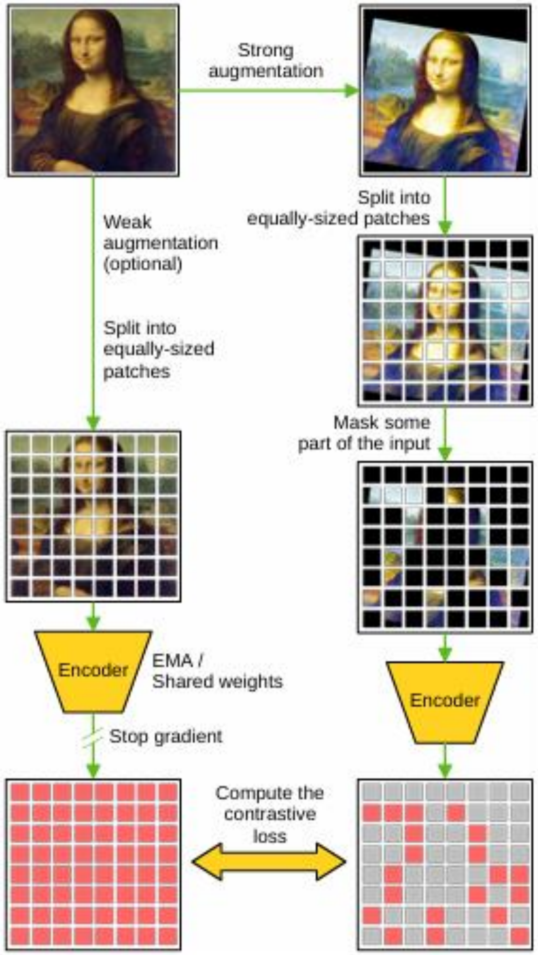
    $\tilde{H}[i] \leftarrow E_\theta(M)$

$\mathcal{L}(\varphi, \theta, M) \leftarrow d(\hat{H}, \tilde{H})$

$\theta \leftarrow \theta - \eta \cdot \frac{\partial \mathcal{L}}{\partial \theta}$

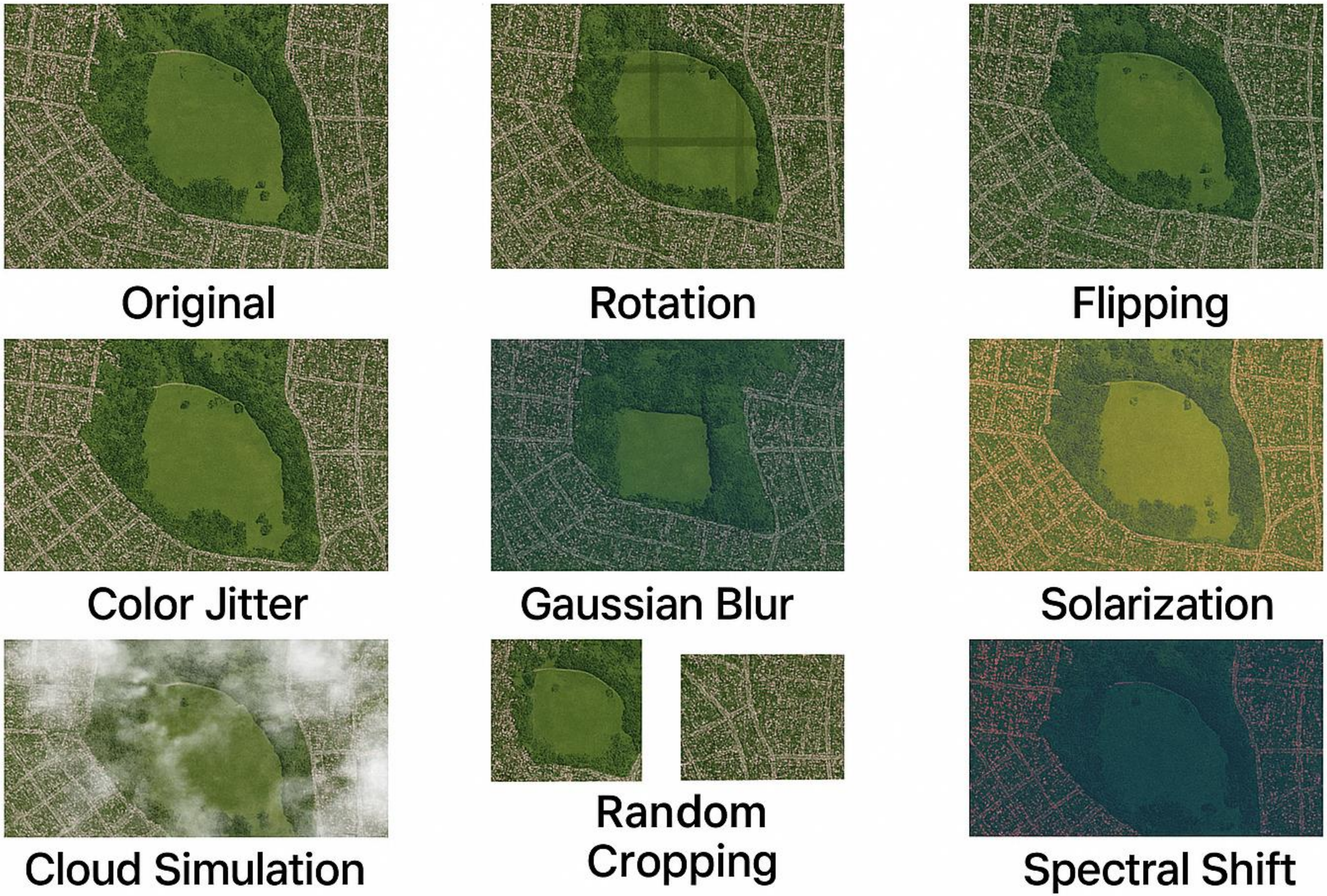
$\varphi \leftarrow \varphi - \eta \cdot \frac{\partial \mathcal{L}}{\partial \varphi}$

$M \leftarrow M - \eta \cdot \frac{\partial \mathcal{L}}{\partial M}$

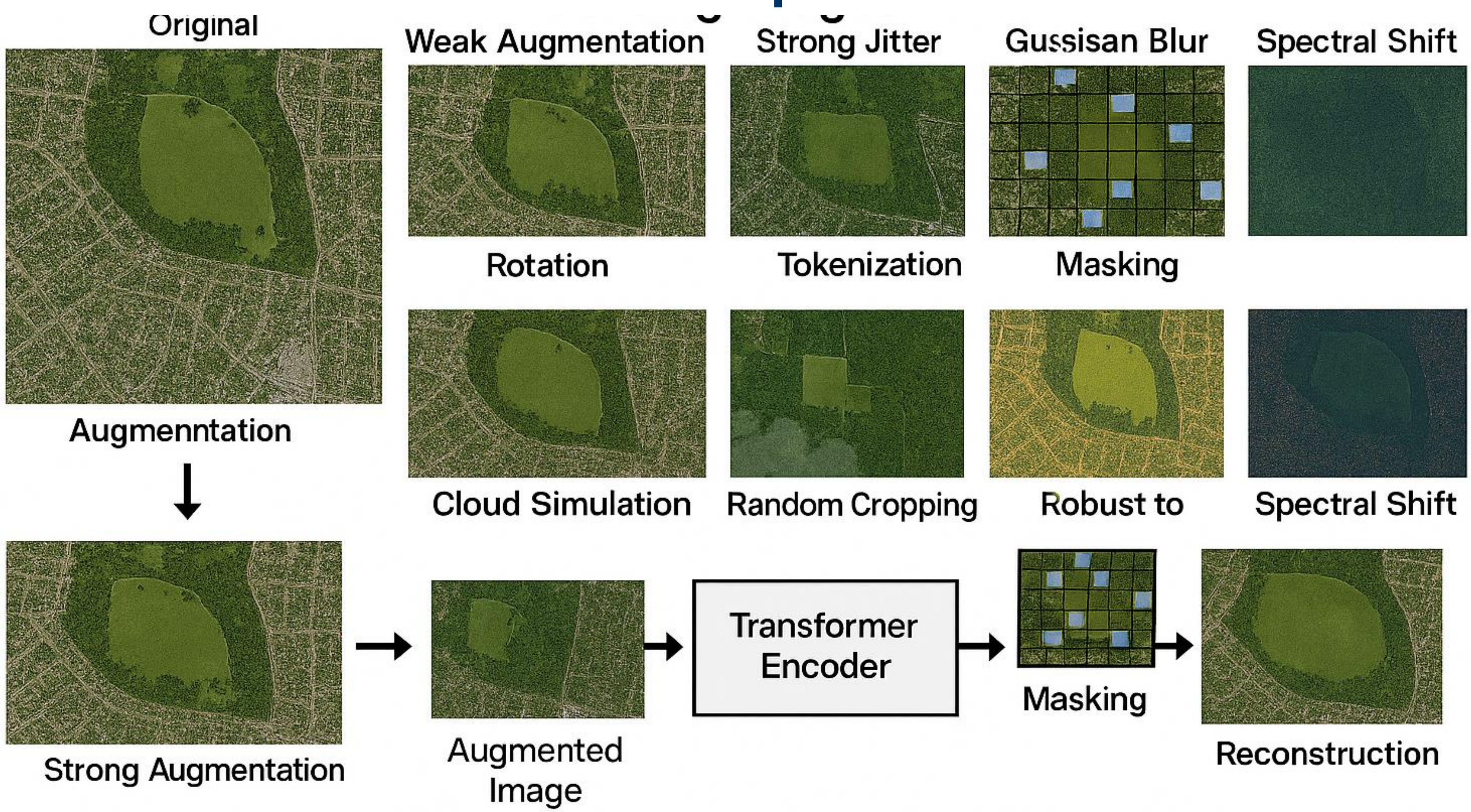


Masked Image Modeling: A Survey ,2025

Augmentation in Remote Sensing



Data Augmentation and Vision Transformer Pipeline for RS



Strong augmentations improve vorusbustness in self-supervised learning for geospatial foundation models like Clay, Prithvi, SatMAE.

Handling the "4th Dimension"

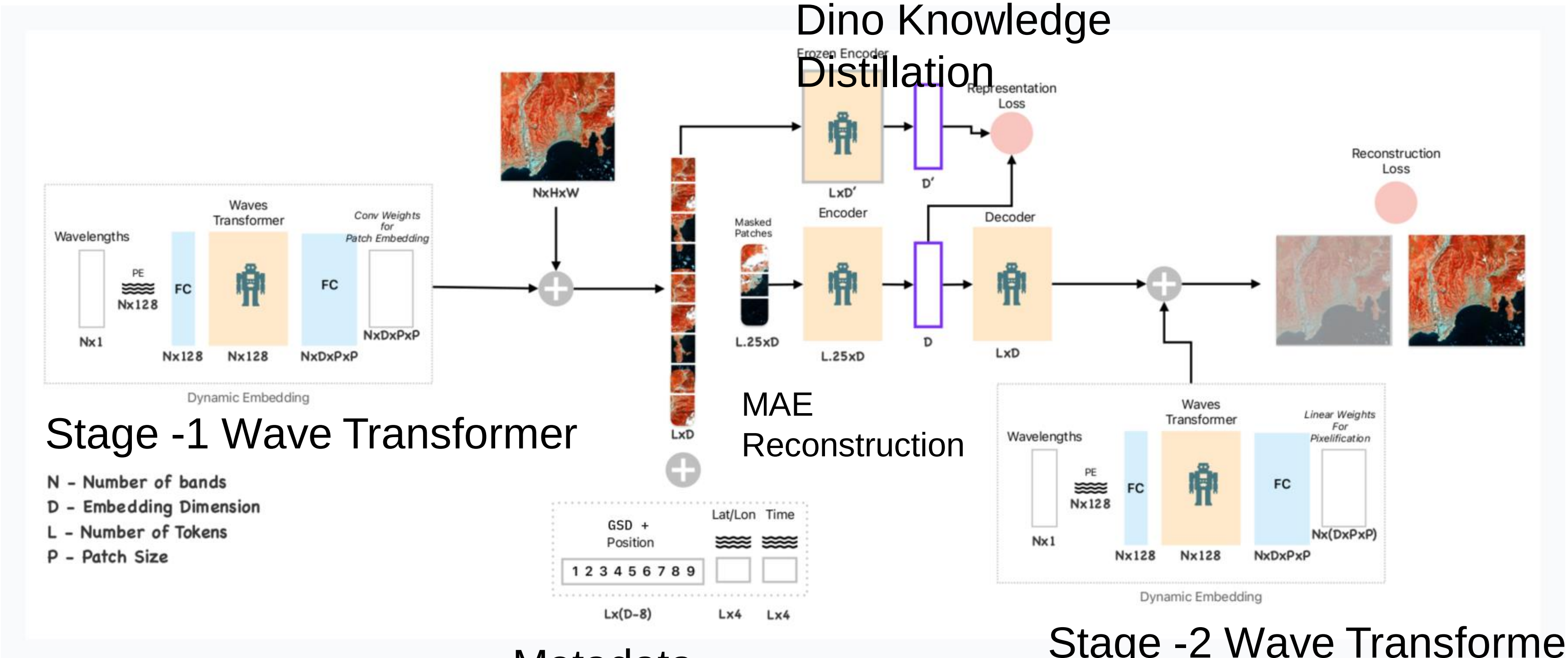
- Satellite images aren't just RGB.
- They have 12+ spectral bands and change over time.
- The 3D Token
  - **Standard ViT:** Patches are 2D squares (x, y).
  - **Geo ViT:** Patches are 3D cubes (x, y, time) or (x, y, spectrum).

Benefit: This allows models to learn that "Green → Brown" over 3 months means "Harvest," not "Deforestation."

Clay Model Architecture

Self-Supervised Dual-Objective Training

Dino Knowledge Distillation



Stage -1 Wave Transformer

MAE Reconstruction

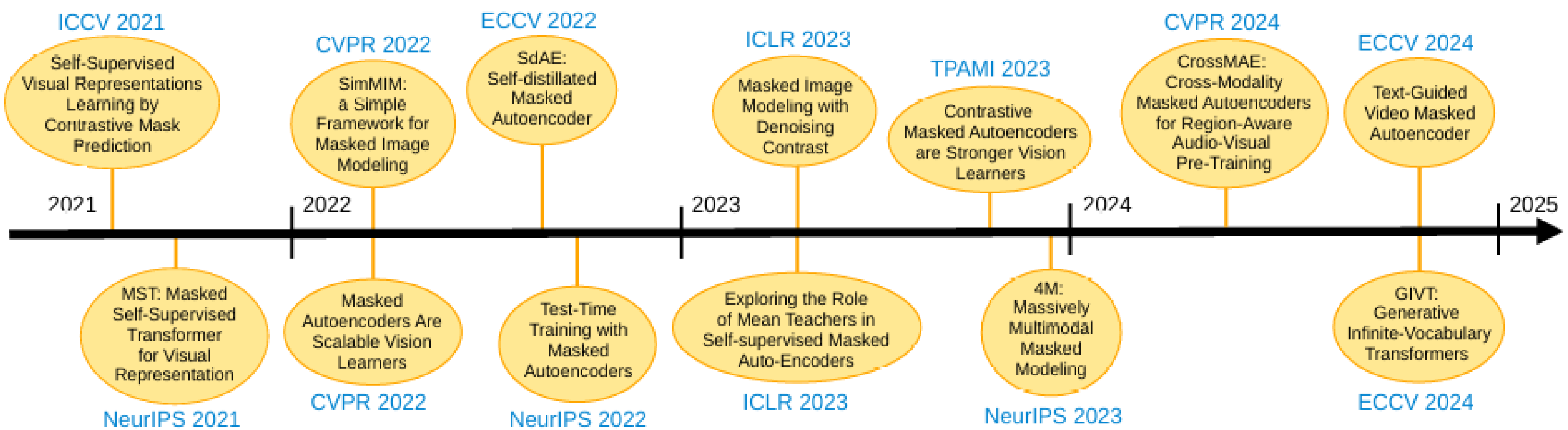
Stage -2 Wave Transformer

Metadata Conditioning

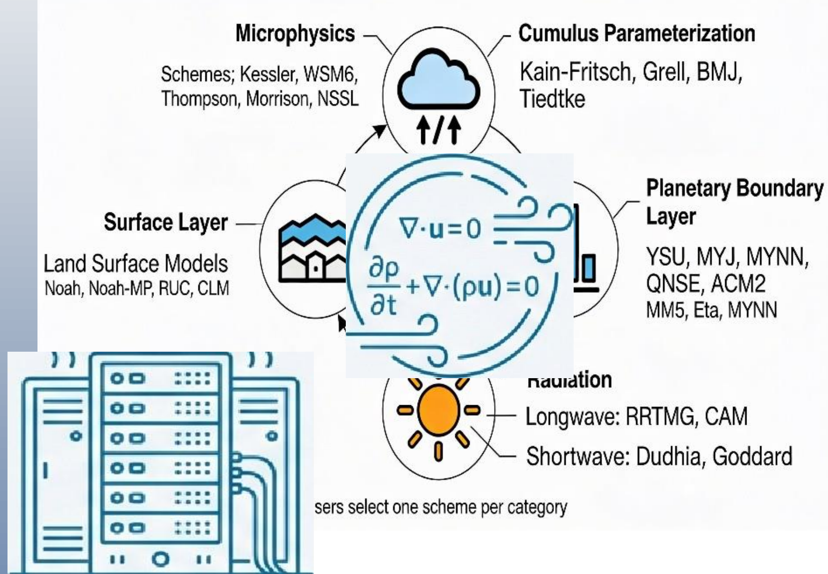
Meta-Learner (Dynamic Embedding Generation)

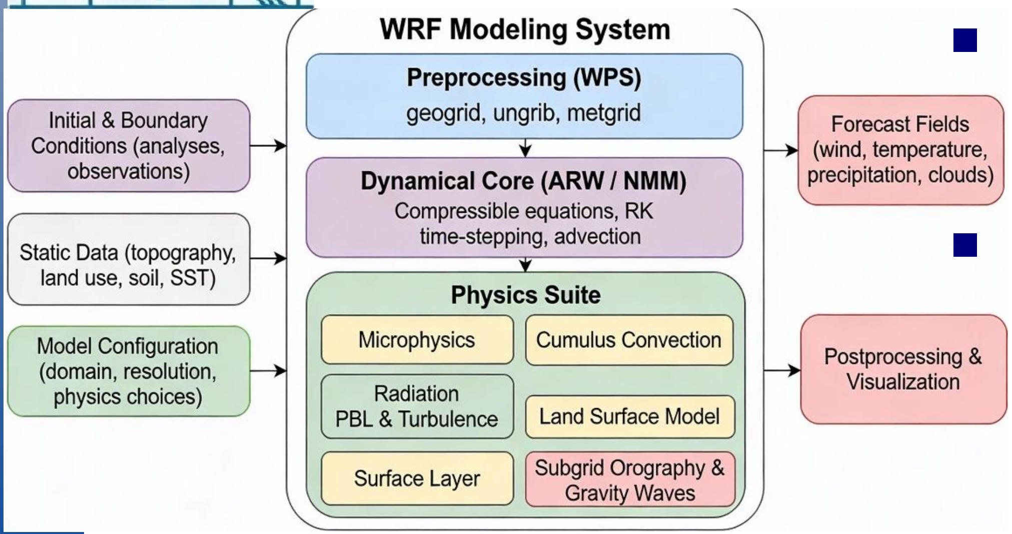
Clay Model

SSL Learning Approaches and Models



Traditional Numerical Weather Prediction (NWP)





- Foundation: The primary method for over 70 years.
- Mechanism: Solves a set of non-linear partial differential equations that govern the physics and dynamics of the atmosphere (conservation of momentum, mass, energy, etc.).
- Process: Simulates atmospheric and ocean conditions on a discretised grid system.
- Demand: Requires immense computational resources and hours to run a single forecast.
- Limitation: Accuracy is constrained by grid resolution and the need to approximate (parameterise) critical processes like cloud microphysics.

The Landscape of Models

- **Temporal** Specialist – Prithvi (IBM/NASA)
  - Temporal Vision Transformer
  - It treats satellite data as a video, not a snapshot.
  - Analyzing changes over time—floods, crop growth cycles, and burn scars.
- **Architecture:** It processes a time series of images simultaneously to understand causality.

The Landscape of Models

- Spectral Specialist – **SatMAE** & **SpectralGPT** Temporal Vision Transformer
  - It treats satellite data as a video, not a snapshot.
  - Analysing changes over time—floods, crop growth cycles, and burn scars.
- Architecture**:** It processes a time series of images simultaneously to understand causality.

The Landscape of Models

- The "Scale" Specialist
  - A house looks different at 30cm resolution vs. 10m resolution.
  - ScaleMAE explicitly encodes the GSD (Ground Sample Distance) into the model.
- You can feed it drone imagery or satellite imagery, and it understands the size of objects relative to the physics of the camera.

The Landscape of Models

- The "Universal" Specialist – DOFA & Clay
- **DOFA (Dynamic One-For-All):** Designed to handle *any* sensor. You can feed it Sentinel-1 (Radar), Sentinel-2 (Optical), or Gaofen data. It adapts dynamically to the number of input channels.
- **Clay:** The open-source standard. Built on a massive dataset to be the "Swiss Army Knife" for Earth. It uses MIM but focuses on being lightweight and easy to fine-tune for environmental tasks..

Summary of Models

Model	Architecture	Best Use Case	Unique Feature
SatMAE	ViT + MAE	General Classification	First to successfully apply MAE to multi-spectral space.
Prithvi	Temporal ViT	Disasters / Agriculture	Understands **Time** (Time-series input).
ScaleMAE	Multi-Scale ViT	Object Detection	Understands **Zoom** (Resolution invariant).
DOFA	Dynamic ViT	Multi-Sensor Fusion	Understands **Different Sensors** (Radar + Optical).
SpectralGPT	3D ViT	Hyperspectral	Understands **Physics** (Deep spectral correlation).
Clay	ViT + MAE	General Purpose	Open, polished, and efficient for diverse tasks.

Thank You

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